

LANGA WORKSPACE — GOOGLE CLOUD ARCHITECTURE

Production Cloud Architecture & Key Lead Engineering Contributions

PORTFOLIO NOTICE & CONFIDENTIALITY ABSTRACT:

This document presents the production Google Cloud Platform (GCP) architecture for Langa Workspace, an enterprise sovereign AI platform powering South African schools. All proprietary business logic, private API endpoints, environment keys, customer PII, and security credentials have been strictly abstracted or omitted for recruiter and technical evaluation use. All documented components reflect real production deployments.

1. System Architecture Diagram

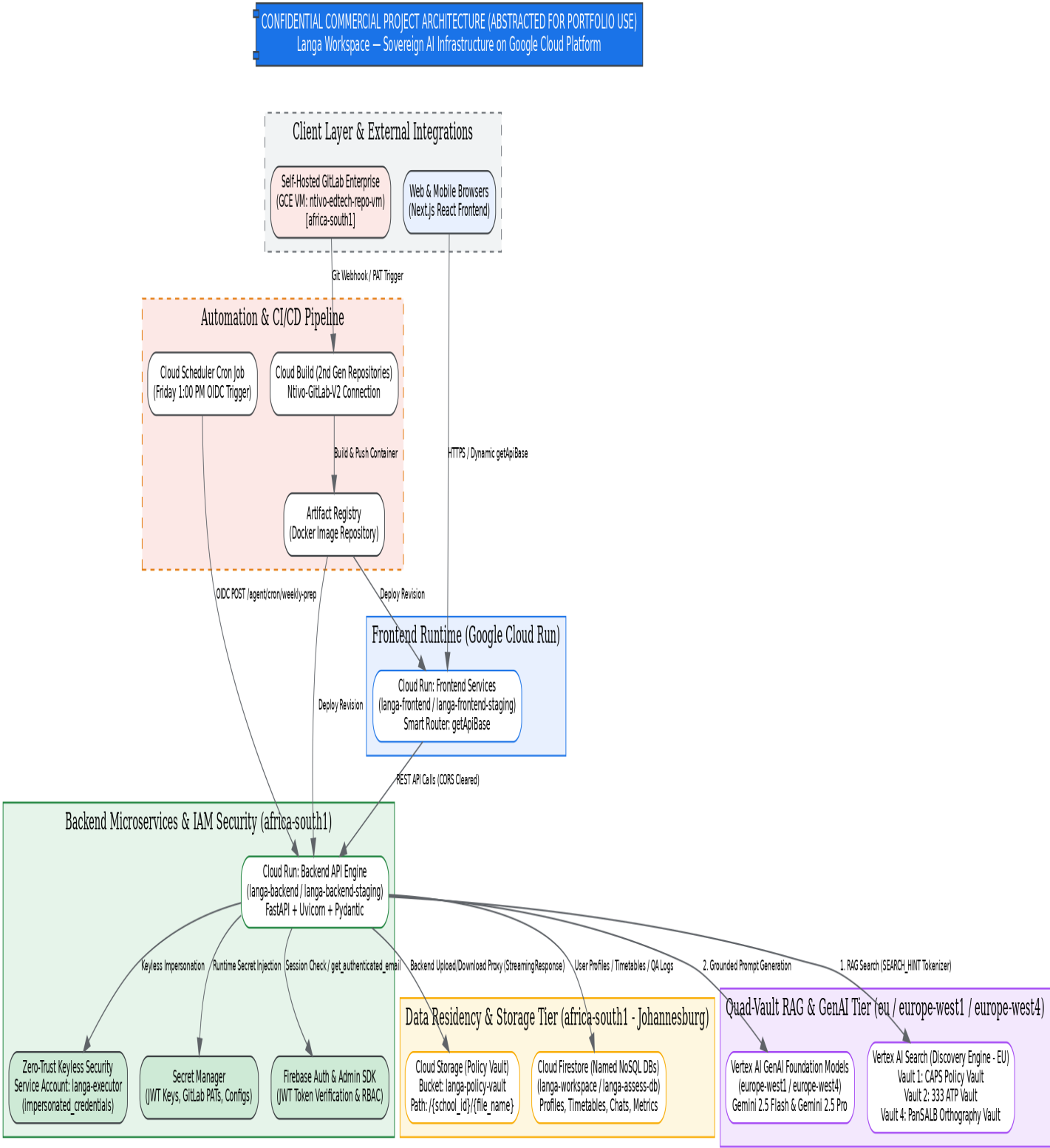


Figure 1.1: Production GCP Topology, RAG Pipelines, Multi-Tenant Boundaries, and CI/CD Automation

2. Documented Lead Engineering Contributions

- **Architectural Decoupling & Serverless Migration:** Migrated legacy monolith into a decoupled, highly responsive microservices stack featuring a Next.js React frontend and a FastAPI backend running on Google Cloud Run in the africa-south1 (Johannesburg) region.
- **Zero-Trust IAM & Keyless Security:** Fortified backend infrastructure by implementing Google Cloud Service Account impersonation via the official impersonated_credentials wrapper (using langa-executor service account), eliminating static private key JSON files from code repositories.
- **Solving the GCP 'Location Paradox':** Solved cross-regional constraints by engineering a dual-client system: keeping POPIA user database records strictly inside Johannesburg (africa-south1) while orchestrating Vertex AI RAG search and Gemini model calls across eu / europe-west1.
- **FastAPI Upload Proxy & Streaming Pipeline:** Replaced brittle GCS client-side signed URLs with a secure FastAPI upload proxy that buffers incoming 10MB+ policy PDFs in-memory and writes to Cloud Storage using native SDKs, while delivering downloads via StreamingResponse with JWT tenancy checks.
- **Smart Runtime Host Router (getApiBase):** Diagnosed and resolved Next.js Docker buildpack environment variable stripping (the /undefined bug) by engineering a runtime browser hostname parser that dynamically targets Localhost, Staging, or Production backends while sanitizing trailing slashes to prevent CORS 307 Redirects.
- **Quad-Vault RAG & Compliance Audit Matrix:** Built a multi-stage RAG execution pipeline integrating CAPS Policy Vault, 333 ATP Vault, and PanSALB Orthography Vault. Engineered a compliance audit matrix that forces the model to calculate and print cognitive mark distributions (30/50/20) prior to assessment drafting.
- **Autonomous Background Worker (Agent Nano):** Engineered an event-driven background worker (agent.py) triggered via Google Cloud Scheduler OIDC calls every Friday at 1:00 PM to pre-draft weekly curriculum lesson plans aligned with South African school calendars and teacher timetables.
- **Multi-Tenant Campus Isolation:** Implemented tenant scope guards and index-free user iteration for HOD and Principal review queues in the Management Hub, avoiding NoSQL index locks while strictly isolating campus records.
- **WeasyPrint LaTeX PDF Engine:** Built a recursive bracket-matching engine (get_matching_brace) to parse deeply nested LaTeX equations into clean HTML/CSS elements for exam paper PDF generation.
- **Freemium Quota Engine & JIT Provisioning:** Implemented Firebase SSO with Just-in-Time Firestore profile creation, a 2-generation/day quota enforcer interceptor, and a blurred paywall upgrade modal.

3. Production GCP Service & Infrastructure Catalog

GCP Component	Deployment Region	Technical Role & Responsibility	Security / Governance Policy
Cloud Run	africa-south1	Serverless microservices engine for Next.js frontend and FastAPI backend containers.	Public invoker with JWT bearer token verification.
Cloud Firestore	africa-south1	Named NoSQL instances (langa-workspace, langa-assess-db) storing user profiles and chats.	Strict POPIA 100% domestic data residency.
Cloud Storage	africa-south1	Object bucket (langa-policy-vault) for heavy policy PDFs (10MB+).	Virtual directory isolation (/{{school_id}}/{{file}}).
Vertex AI Search	eu (Multi-region)	Discovery Engine datastores: CAPS Policy, 333 ATPs, and PanSALB Orthography.	Curriculum-as-Code RAG retrieval engine.
Vertex AI Models	europe-west1 / west4	Foundation LLMs (gemini-2.5-flash and gemini-2.5-pro) for content drafting.	Zero-Learner-Data policy; isolated from training.

GCP Component	Deployment Region	Technical Role & Responsibility	Security / Governance Policy
Cloud Scheduler	africa-south1	Cron heartbeat issuing OIDC calls every Friday at 1:00 PM for Agent Nano.	X-Cloudscheduler-Jobname authentication.
Cloud Build	Global / Host	2nd Gen Repositories (Ntivo-GitLab-V2) CI/CD build automation pipeline.	Authenticated via GitLab PAT tokens in Secret Mgr.
Artifact Registry	africa-south1	Docker image repository storing compiled container images.	IAM access control for Cloud Run deployments.
Secret Manager	africa-south1	Stores environment credentials, JWT signing secrets, and API access tokens.	Injected dynamically into Cloud Run containers.
IAM Security	Global	Keyless impersonation via langa-executor service account wrapper.	Zero-Trust Service Account Impersonation.

4. End-to-End Data Flow & Request Lifecycle

1. **1. Client Request & Dynamic Routing:** Educator initiates a generation request on the Next.js React frontend. The runtime getApiBase parser evaluates window.location.hostname, sanitizes trailing slashes, and routes the request to the Cloud Run FastAPI backend.
1. **2. Security & Session Interception:** FastAPI's get_authenticated_email dependency intercepts the Authorization: Bearer <token> JWT, verifies the signature using Firebase Admin SDK, and extracts the cryptographically verified user identity and role.
1. **3. RAG Retrieval (The Librarian Phase):** The backend tokenizer generates a structured SEARCH_HINT and queries Vertex AI Search Discovery Engine in the eu multi-region to retrieve exact CAPS policy rules, ATP week pacing, and PanSALB orthography constraints.
1. **4. Model Generation (The Brain Phase):** The retrieved context is bundled into a system prompt and sent to Gemini 2.5 Flash/Pro in europe-west1. The model enforces the Compliance Audit Matrix (30/50/20 cognitive split) before generating output.
1. **5. Storage & Telemetry:** Generated content is stored in Cloud Firestore (africa-south1) under the user's isolated document path, automatically updating hours reclaimed metrics via real-time Firebase listeners.